

Structural and Strongly Structural Input and State Observability of Linear Network Systems

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Abstract—This paper studies linear network systems affected by multiple unknown inputs with the objective of reconstructing both the initial state and the unknown input with one time-step delay. We state conditions under which both the whole network state and the unknown input can be reconstructed from output measurements, over every window of length N , with N being the dimension of the system, for all system matrices that share a common zero/nonzero pattern (uniform N -step strongly structural input and state observability) or at least for almost all system matrices that share a common zero/nonzero pattern (uniform N -step structural input and state observability). Based on some specific assumptions on the structure of the interactions between the unknown input and the network states, we show that such a characterization depends only on strongly structural (respectively, structural) observability properties of a suitable subsystem.

Index Terms—Cyber-physical security, input and state observability (ISO), linear network systems, strongly structural observability, structural observability.

I. INTRODUCTION

THE FIELDS of application of network systems span from critical infrastructure domains such as power networks, water and gas distribution networks to healthcare systems and flight control systems, among others. Given the ubiquitous nature of their usage, it is of paramount importance to ensure that each individual subsystem functions as desired. The notion of *observability* enables one to achieve this by expending as few resources as possible.

However, such systems are quite likely to malfunction due to *local* attacks by malicious agents modeled as external unknown inputs [1], which could have significant consequences as evidenced by the failure of wastewater-management systems in Marochy Australia, in early 2000 [2] and multiple power black-outs in Brazil [3] to cite a few. Hence, in addition to being able to observe the state in the presence of unknown inputs (also known as *strong observability* [4], [5]) it is crucial that the unknown

input be observed as well. This notion is known as *input and state observability*, which hereafter is referred to as *ISO*. In the context of linear time varying (LTV) systems, a system being ISO over an interval does not necessarily mean that the mentioned system would be ISO over every sufficiently long interval. The concept of *uniform δ -step ISO* (i.e., ISO over every time window of length δ) gets rid of this drawback.

The notion of ISO is of particular importance in designing unbiased minimum-variance filters that *simultaneously* estimate both state and unknown inputs [6]–[8]. It is well known that algebraic approaches toward characterizing ISO involve the classic Kalman-like rank condition or a variant of the Popov–Belevitch–Hautus (PBH) test. (See, e.g., [9] and [8]). Both tests require exact knowledge of entries in the matrices of interest and are computationally heavy as the dimension of the system grows, while the latter is not suitable for LTV systems. This leads to the study of ISO based on the structure of the underlying network (represented by a graph) and the corresponding line of work is known as *structured systems*.

We say that a linear system is *structured* if the system matrices have coefficients that are either a fixed zero or a free parameter (i.e., the coefficients may take any value in $\mathbb{R}$). Thus, one can capture a family of systems that obey a certain rule. Under such a setup, if a property holds for *almost all* choices of entries in the nonzero positions of the system matrices, it is called *structural* [10], whereas if it holds for *all nonzero* choices of entries in the nonzero positions of the system matrices, it is called *strongly structural* [11] (s-structural).

For linear time-invariant (LTI) systems, structural controllability or the dual notion of observability has been studied since [10], while graph-theoretic characterizations for s-structural controllability were first provided in [11]. The survey paper [12] revises some graph-theoretic characterizations for observability in addition to recalling similar conditions for solvability issues like disturbance rejection, input–output decoupling, and so on, while equivalent characterizations for s-structural controllability have been provided in [13]. More recently, for s-structural controllability, [14] provides necessary and sufficient conditions in terms of *uniquely restricted matching* (precise definition appears in Section VI), while [15] does so in terms of *zero forcing sets*.

Notice that all of the results mentioned as yet are for LTI systems, where both parameters and the structure remain constant over time. In contrast, it is natural to assume that the parameters can evolve over time while the structure remains fixed (LTV with fixed topology). Under such a scenario, necessary and

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sufficient conditions for structural observability of LTV systems are given in [16], while necessary and sufficient conditions for s-structural observability are available in [17]. However, these results are not applicable for LTV systems with *unknown* inputs.

To the best of our knowledge, for discrete-time linear structured systems, a graph-theoretic characterization for the more general ISO problem encompassing *multiple* unknown inputs and accounting for LTV dynamics is missing. For continuous-time LTI systems, [18] gives necessary and sufficient conditions for structural ISO, which, when translated into a discrete-time setup, yields structural ISO with *some* delay L . This paper deals with the notion of *ISO with delay-1*, a notion that is essential for running input and state estimation filters. For discrete-time LTV systems, [19] gives a characterization of s-structural ISO *but not of structural ISO*, while an earlier work from the authors gives characterizations of both structural and s-structural ISO for LTI systems affected by a *single* unknown input [20]. As such, the main contributions of this paper are threefold; under suitable assumptions on the structure of the input and output matrices, first, we show equivalence between ISO of a linear system and the observability of a suitably defined subsystem. Second, we give a characterization of uniform N -step structural (see Theorem 1) [respectively, uniform N -step strongly structural (see Theorem 2)] ISO, that is, the conditions under which both the whole network state and the unknown input can be reconstructed for *almost all* (respectively, *all*) system matrices that share a common zero/nonzero pattern, over every time window of length N . This equivalence enables one to study structural (respectively, s-structural) ISO using the graph techniques given in [21] and [14] (respectively, [14] and [15]).

The organization of this paper is as follows. We state the problem in Section II. Section III presents an algebraic characterization of the ISO problem, while Section IV, under suitable assumptions on the input and output matrices, shows the equivalence between ISO and observability of an appropriate subsystem. Section V discusses structural ISO, while Section VI studies the stronger notion of s-structural ISO. Different from Sections V and VI, Section VII explores structural and s-structural ISO without assumptions on input and output matrices, while Section VIII presents concluding remarks along with discussing future lines of research.

Notations: $\mathbb{R}$, $\mathbb{R}_*$, and $\mathbb{Z}$ denote the set of real numbers, nonzero real numbers, and integers, respectively. $e_{j,N}$ represents the j th vector of the canonical basis of $\mathbb{R}^N$. Alternatively, assuming that the length is clear from context, we would denote the same as just e_j . $[A]_{i,j}$ denotes the entry in matrix A that corresponds to its i th row and j th column. I_N denotes an identity matrix of size N . Given two matrices A and B , let $A \odot B$ and $A \otimes B$ denote the entrywise product and Kronecker product, respectively. $A = \text{diag}(A_1, A_2, \dots, A_N)$ denotes a block diagonal matrix whose blocks along the diagonal are $A_1, A_2, \dots, A_n$. In case $A_1 = A_2 = \dots = A_N$, we get $\text{diag}(A_1, A_2, \dots, A_N) = I_N \otimes A$. $\{A_k\}_{k_0}^{k_1}$ denotes a sequence of matrices A_k , $k = k_0, k_0 + 1, \dots, k_1$. $|\mathcal{X}|$ denotes cardinality of a set $\mathcal{X}$. $\lceil a \rceil$ denotes the smallest integer greater than or equal to a .

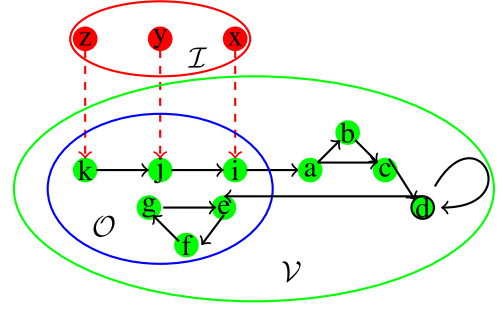


Fig. 1. Graph representation of a network system affected by external agents where $\mathcal{I}$, $\mathcal{O}$, and $\mathcal{V}$ are the sets of attackers, observed nodes, and state nodes, respectively, while the set of assailable nodes is $\mathcal{A} = \{i, j, k\}$.

II. PROBLEM STATEMENT

Consider a linear network system with N nodes, represented by a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ is the vertex set and $\mathcal{E} = \{(j, i) \in \mathcal{V} \times \mathcal{V} \mid [A_{\mathcal{G}}]_{i,j} = 1\}$; $A_{\mathcal{G}}$ is the adjacency matrix of $\mathcal{G}$. In this network, some states can be directly measured. They define the set $\mathcal{O} = \{j_1, j_2, \dots, j_M\} \subseteq \mathcal{V}$, with M being the number of observed states. From an analysis of the network, we assume that $\mathcal{V}$ can also be partitioned into assailable nodes and reliable ones. We define by $\mathcal{A} = \{i_1, i_2, \dots, i_R\}$, the set of the R assailable nodes that *may* be attacked by P external malicious agents defining the set denoted by $\mathcal{I}$, the attack being modeled as an unknown input. An illustration is shown in Fig. 1, where three malicious nodes, namely, x, y, z can attack the network with vertex set $\mathcal{V}$ through agents k, j , and i . A setup of this sort can be used as an abstraction to model attacks on multiple nodes including deception attacks [22], false data injection [23], fault diagnosis and detection [24], and input estimation in physiological systems [25].

The dynamics of the linear network system described above is given by the following equations:

$$\begin{aligned} x_{k+1} &= W_k x_k + A_B u_k \\ y_k &= A_C x_k \end{aligned} \quad (1)$$

with state vector $x_k \in \mathbb{R}^N$, unknown input vector $u_k \in \mathbb{R}^P$, and output vector $y_k \in \mathbb{R}^M$. Furthermore, $W_k \in \mathbb{R}^{N \times N}$, $A_B \in \mathbb{R}^{N \times P}$, and $A_C \in \mathbb{R}^{M \times N}$, where W_k , A_B , and A_C are state matrix at time instant k , input matrix, and output matrix, respectively. For the rest of this paper, over a given interval $[k_0, k_1]$, we denote the dynamics of an LTV system as $\{W_k, A_B, A_C\}_{k_0}^{k_1}$, while that of an LTI system is indicated by $\{W, A_B, A_C\}_{k_0}^{k_1}$.

In what follows, we assume that for all k belonging to $\mathbb{Z}$, $W_k \in \mathcal{W}$ or, in particular, $W_k \in \mathcal{W}^*$, where matrix set $\mathcal{W} = \{Z_1 \odot A_{\mathcal{G}} \mid Z_1 \in \mathbb{R}^{N \times N}\}$ and $\mathcal{W}^* = \{Z_1 \odot A_{\mathcal{G}} \mid Z_1 \in \mathbb{R}_*^{N \times N}\}$. Both $\{\mathcal{W}, A_B, A_C\}$ and $\{\mathcal{W}^*, A_B, A_C\}$ impose a fixed zero structure. The remaining coefficients (i.e., *not fixed to zero* positions) of the matrices are referred to as *free parameters*. The free parameters of matrices belonging to $\mathcal{W}$ can take *any* values, while those belonging to $\mathcal{W}^*$ strictly take *nonzero*

values. Notice that this assumption implies that the topology of $\mathcal{G}$ remains fixed but the entries corresponding to the free parameters of the system matrices may vary. For the particular case of LTI systems, the mentioned entries remain constant.

We narrow our attention to the case wherein each unknown input affects exactly one node of $\mathcal{G}$ and each node is, at most, affected by a single unknown input. This leads to the following assumption:

Assumption 1 (A1):

$$A_B = \begin{bmatrix} e_{i_1;N} & e_{i_2;N} & \cdot & \cdot & e_{i_P;N} \end{bmatrix},$$

$$A_C^T = \begin{bmatrix} e_{j_1;N} & e_{j_2;N} & \cdot & \cdot & e_{j_M;N} \end{bmatrix}.$$

In the context of network systems, it is natural to think of states as local variables that are in different physical locations, whereas unknown inputs could be isolated entities that are, at best, able to attack a single state. For instance, the topology of a power distribution network can be considered as the connectivity among the meters installed at the substation, feeders, transformers, and consumer mains. An attack corresponds to addition or draining of active power, while the state at each node can be measured using smart meters.

As a consequence of assumption A1, we rule out scenarios wherein a linear combination of multiple unknown inputs affects a single node in $\mathcal{G}$. Therefore, we have $R = P$. On the other hand, the unknown inputs are of arbitrary nature, and for the particular case in which some of the unknown inputs are the same, we would have a single unknown input affecting multiple nodes in $\mathcal{G}$ and, as such, we provide *sufficient but not necessary* conditions for this setup as well.

In this paper, we first study conditions under which it is possible to jointly estimate both the initial state and the sequence of multiple unknown inputs for an LTV system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ as well as the particular case of an LTI system $\{W, A_B, A_C\}_{k_0}^{k_1}$ from measurements of a subset of state vertices. Thereafter, based only on the structure of the graph $\mathcal{G}$, we will characterize ISO for 1) *almost all* choices of entries in $\mathcal{W}$ (see Section V) and 2) *every choice of entries* in $\mathcal{W}^*$ (see Section VI), over all sufficiently long time windows.

III. ALGEBRAIC CHARACTERIZATION

In this section, we describe some algebraic criteria for observability and for ISO, reviewing the relevant classical results together with some new variation of them concerning ISO, see, in particular, Proposition 4.

A. Definitions

The concept of observability was first introduced by Kalman in his seminal paper [26]. We recall it in the following definition.

Definition 1: The system $\{W_k, A_C\}_{k_0}^{k_1}$ is *observable on* $[k_0, k_1]$ if any initial state x_{k_0} is uniquely determined by the corresponding measured output sequence $\{y_{k_0}, y_{k_0+1}, \dots, y_{k_1}\}$. ■

It is worth noticing here that Definition 1 explicitly asks that the initial state x_{k_0} be reconstructed, assuming that the

input is *known*. On the other hand, the notion of strong observability asks that the initial state x_{k_0} be reconstructed even in the presence of an *unknown* input, while that of *left invertibility* with delay 1 requires that the inputs from u_{k_0} up to u_{k_1-1} be reconstructed from the outputs up to y_{k_1} . These two notions, namely, strong observability and left invertibility with delay 1, give rise to the definition of *ISO*.

Definition 2: The system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO, with delay 1, on the interval $[k_0, k_1]$ if the initial condition $x_{k_0} \in \mathbb{R}^N$ and the unknown inputs sequence $\{u_{k_0}, u_{k_0+1}, \dots, u_{k_1-1}\}$ can be uniquely determined from the measured output sequence $\{y_{k_0}, y_{k_0+1}, \dots, y_{k_1}\}$. ■

A stronger notion of observability is that of *uniform δ -step observability*, which requires that a system be observable over every time window of length δ [27]. Analogously, we define *uniform δ -step ISO* as follows.

Definition 3: The system $\{W_k, A_B, A_C\}_{k \in \mathbb{Z}}$ is uniformly δ -step ISO if $\forall k_0 \in \mathbb{Z}$ $\{W_k, A_B, A_C\}_{k_0}^{k_0+\delta}$ is ISO over $[k_0, k_0 + \delta]$. ■

Remark 1: Notice that although uniform δ -step ISO (respectively, observability) is with respect to all intervals of length δ , it turns out that it can be rephrased considering all intervals of length *at least* δ . For observability, this is immediate: if a system is observable over $[k_0, k_0 + \delta]$, then it is also observable over $[k_0, k_0 + \eta]$ for all $\eta \geq \delta$. For ISO, one needs to reconstruct all inputs up to $k_0 + \eta - 1$ and not only those up to $k_0 + \delta - 1$. If the system is uniformly δ -step observable, it is possible to use δ -step ISO over successive time windows of length δ to ensure that all of the required inputs are indeed reconstructed. ■

Remark 2: It is well known that either LTI systems are not observable or are uniformly N -step observable, in which case, we would simply call it *observable*. ■

B. Observability, Invertibility, and Input and State Observability Matrices

Let Θ_{k_0, k_1} , Γ_{k_0, k_1} , and Ψ_{k_0, k_1} represent the observability matrix, invertibility matrix, and ISO matrix, respectively, over the interval $[k_0, k_1]$. These are defined as follows:

$$\Theta_{k_0, k_1} = \begin{bmatrix} A_C \\ A_C W_{k_0} \\ A_C W_{k_0+1} W_{k_0} \\ \cdot \\ \cdot \\ \cdot \\ A_C W_{k_1-1} \cdots W_{k_0} \end{bmatrix},$$

$$\Gamma_{k_0, k_1} = \begin{bmatrix} 0 & \cdots & \cdots & 0 \\ A_C A_B & \cdots & \cdots & 0 \\ A_C W_{k_0+1} A_B & A_C A_B & \cdots & 0 \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ A_C W_{k_1-1} \cdots W_{k_0+1} A_B & \cdots & \cdots & A_C A_B \end{bmatrix},$$

$$\Psi_{k_0, k_1} = [\Theta_{k_0, k_1} \Gamma_{k_0, k_1}]. \quad (2)$$

This leads us to the following classical Kalman-like characterization of ISO.

Proposition 1: The system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO over $[k_0, k_1]$ if and only if Ψ_{k_0, k_1} is full-column rank, that is, $\text{rank}(\Psi_{k_0, k_1}) = N + (k_1 - k_0)P$. ■

Proof: Let $y_{k_0:k_1}$ and $u_{k_0:k_1-1}$ denote the vectors of concatenated outputs and unknown inputs over $[k_0, k_1]$, respectively. Therefore, from (1) and (2), the following can be readily obtained:

$$y_{k_0:k_1} = \Theta_{k_0, k_1} x_{k_0} + \Gamma_{k_0, k_1} u_{k_0:k_1-1} = \Psi_{k_0, k_1} \begin{bmatrix} x_{k_0} \\ u_{k_0:k_1-1} \end{bmatrix}.$$

Based on Definition 2, it is immediate that ISO is equivalent to uniqueness of the above-mentioned system of linear equations. ■

Proposition 1 enables one to exploit the structure of Ψ_{k_0, k_1} so as to find some simple necessary conditions for Ψ_{k_0, k_1} to have full-column rank. The following proposition briefly summarizes them.

Proposition 2: The following conditions are necessary for the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ to be ISO over $[k_0, k_1]$:

- i) $\text{rank}(\Theta_{k_0, k_1}) = N$,
- ii) $\text{rank}(A_C A_B) = P$,
- iii) $M \geq P$,
- iv) $N \geq P$.

In case $N > P$, then the following conditions are also necessary:

- v) $M > P$,
- vi) $k_1 - k_0 \geq \lceil \frac{N-M}{M-P} \rceil$.

In case $P = N$, then the following conditions are necessary and sufficient:

$M = N \quad \forall k \in [k_0, k_1] \text{ rank}(A_C) = N \text{ and } \forall k \in [k_0, k_1 - 1] \text{ rank}(A_B) = N$. ■

Proof: The proof is reported in the Appendix. ■

Notice that Proposition 2 fully characterizes the ISO problem for the particular case of $P = N$, where, under A1, the system is ISO if and only if all nodes are observed (i.e., $\mathcal{O} = \mathcal{V}$). In this paper, we restrict our attention to the nontrivial case of $N > P$, that is, *not all* the nodes are assailable. Therefore, from Proposition 2, $M > P$ is a necessary condition for ISO.

From Proposition 2, we know that the following are conditions necessary for ISO:

- 1) all of the assailable nodes are observed, that is, $\{i_1, i_2, \dots, i_P\} \subset \{j_1, j_2, \dots, j_M\}$ and
- 2) all of the assailable nodes are distinct, that is, there does not exist h, k belonging to $\{1, 2, \dots, P\}$ such that $i_h = i_k$.

We also assume that all of the observed nodes are distinct, that is, there does not exist h, k belonging to $\{1, 2, \dots, M\}$ such that $j_h = j_k$. This ensures that there are no repeated or dependent rows in C .

Therefore, one can relabel the nodes in $\mathcal{G}$ in the following manner: $i_1 = j_1 = 1, i_2 = j_2 = 2, \dots, i_P = j_P = P$.

The aforesaid relabeling allows us to rewrite A_B and A_C as follows:

Assumption 2 (A2):

$$A_B = \begin{bmatrix} e_{1;N} & e_{2;N} & \dots & e_{P;N} \end{bmatrix},$$

$$A_C^T = \begin{bmatrix} e_{1;N} & e_{2;N} & \dots & e_{P;N} & e_{j_{P+1};N} & \dots & e_{j_M;N} \end{bmatrix}. \quad \blacksquare$$

C. Alternative Algebraic Characterization

Proposition 1 characterizes ISO in terms of rank of Ψ_{k_0, k_1} . However, the elements in Ψ_{k_0, k_1} are obtained by taking products of the state matrices over the interval $[k_0, k_1]$. Consequently, the zero/nonzero pattern is not preserved. In order to overcome this drawback, in this section, we provide an alternative algebraic characterization for both observability and ISO.

Reference 28, Theorem 6.4.1, gives an alternative characterization of controllability. The following proposition does the same for observability.

Proposition 3: The system $\{W_k, A_C\}_{k_0}^{k_1}$ is observable over $[k_0, k_1]$ if and only if $\text{rank}(\mathcal{Q}_{k_0, k_1}) = (k_1 - k_0 + 1)N$, where

$$\mathcal{Q}_{k_0, k_1} = \begin{bmatrix} A_C & 0 & \dots & \dots & 0 \\ 0 & A_C & \dots & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & A_C \\ W_{k_0} & -I_N & \dots & \dots & 0 \\ 0 & W_{k_0+1} & -I_N & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & W_{k_1-1} & -I_N \end{bmatrix}$$

with $\mathcal{Q}_{k_0, k_1} \in \mathbb{R}^{(k_1 - k_0 + 1)M + (k_1 - k_0)N \times (k_1 - k_0 + 1)N}$. ■

Proof: Notice that the problem of reconstructing x_{k_0} from $y_{k_0:k_1}$ is equivalent to the problem of reconstructing $x_{k_0}, x_{k_0+1}, \dots, x_{k_1}$. The relationship between the states and outputs can be expressed via a system of linear equations as follows.

From (1) and setting $u(k) = \mathbf{0}_P$, we have $\forall k \in [k_0, k_1 - 1]$, $W_k x_k - x_{k+1} = \mathbf{0}_N$, and $\forall k \in [k_0, k_1] C_k x_k = y_k$.

This can be rewritten as $\mathcal{Q}_{k_0, k_1} x_{k_0:k_1} = \begin{bmatrix} y_{k_0:k_1} \\ \mathbf{0}_{(k_1 - k_0 - 1)N} \end{bmatrix}$.

Hence, the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is observable, over $[k_0, k_1]$, if and only if the aforementioned system of linear equations has a unique solution. ■

It turns out that similar arguments can be made for ISO as well and will be shown in Proposition 4. As a first step, we define the following matrix:

$$\mathcal{J}_{k_0, k_1} = \begin{bmatrix} 0 & \mathcal{Q}_{k_0, k_1} \\ \mathbf{B}_{k_0, k_1} & \end{bmatrix}$$

where $\mathbf{B}_{k_0, k_1} = I_{k_1 - k_0} \otimes A_B \in \mathbb{R}^{(k_1 - k_0)N \times (k_1 - k_0)P}$.

Proposition 4: The system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO over $[k_0, k_1]$ if and only if $\text{rank}(\mathcal{J}_{k_0, k_1}) = (k_1 - k_0)P + (k_1 - k_0 + 1)N$. ■

Proof: The problem of reconstructing x_{k_0} and $u_{k_0:k_1-1}$ from $y_{k_0:k_1}$ is equivalent to the problem of reconstructing $x_{k_0}, x_{k_0+1}, \dots, x_{k_1}$ and $u_{k_0:k_1-1}$. From (1), we have $\forall k \in [k_0, k_1 - 1]$, $W_k x_k + A_B u_k - x_{k+1} = \mathbf{0}_N$ and $\forall k \in [k_0, k_1]$ $A_C x_k = y_k$. Hence, both the state equation and output equation at each time instant can be expressed as a linear combination of $x_{k_0}, x_{k_0+1}, \dots, x_{k_1}$ as well as $u_{k_0}, u_{k_0+1}, \dots, u_{k_1-1}$, in the following manner:

$$\mathcal{J}_{k_0, k_1} \begin{bmatrix} u_{k_0:k_1-1} \\ x_{k_0:k_1} \end{bmatrix} = \begin{bmatrix} y_{k_0:k_1} \\ \mathbf{0}_{(k_1-k_0)N} \end{bmatrix}.$$

Hence, the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO over $[k_0, k_1]$ if and only if the aforementioned system of linear equations has a unique solution. ■

IV. ISO AS OBSERVABILITY OF AN APPROPRIATE SUBSYSTEM

The objective here is to decompose the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ into two subsystems and show that ISO is equivalent to observability of one of the subsystems.

It is crucial to notice here that the identity of the nodes being assailable remains fixed and, according to assumption A2, is equal to $\{1, 2, \dots, P\}$. Consequently, the nodes labeled from $i_{P+1}, \dots, i_N$ are *not assailable*. This enables us to decompose the state vector in two blocks: $\hat{x}_k$ denoting states that are *directly* affected by the unknown inputs and $\tilde{x}_k$ for the remaining states; a corresponding partitioning is also done for the output vector, obtaining

$$x_k = \begin{bmatrix} \hat{x}_k \\ \tilde{x}_k \end{bmatrix}, \quad y_k = \begin{bmatrix} \hat{y}_k \\ \tilde{y}_k \end{bmatrix}$$

with $\hat{x}_k \in \mathbb{R}^P$, $\tilde{x}_k \in \mathbb{R}^{N-P}$, $\hat{y}_k \in \mathbb{R}^P$, and $\tilde{y}_k \in \mathbb{R}^{M-P}$. Moreover, thanks to assumption A2, the input and output matrices can be rewritten as follows:

$$A_B = \begin{bmatrix} I_P \\ \mathbf{0} \end{bmatrix}, \quad A_C = \begin{bmatrix} I_P & \mathbf{0} \\ \mathbf{0} & \tilde{A}_C \end{bmatrix}.$$

Therefore, the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ can be decomposed into two subsystems as follows:

$$\begin{cases} \hat{x}_{k+1} = \hat{W}_k \hat{x}_k + \Lambda \tilde{x}_k + u_k \\ \hat{y}_k = \hat{x}_k \end{cases} \quad (3)$$

$$\begin{cases} \tilde{x}_{k+1} = \tilde{W}_k \tilde{x}_k + \Omega \hat{x}_k \\ \tilde{y}_k = \tilde{A}_C \tilde{x}_k \end{cases} \quad (4)$$

where we use the notation

$$W_k = \begin{bmatrix} \hat{W}_k & \Lambda_k \\ \Omega_k & \tilde{W}_k \end{bmatrix}.$$

From (3), it is clear that $\hat{x}_k$ is directly observed. Hence, (3) represents a system with *known state but two unknown inputs*, namely, $\tilde{x}_k$ and u_k , while (4) represents a system with *unknown state but known input*. Hence, we have the following proposition.

Proposition 5: Under A2, the system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO over $[k_0, k_1]$ if and only if the system $\{\tilde{W}_k, \tilde{A}_C\}_{k_0}^{k_1}$ is observable over $[k_0, k_1]$. ■

Proof: We define the matrices Q_N and $\bar{Q}_N$ as follows:

$$Q_N = \begin{bmatrix} I_P \\ 0 \end{bmatrix}, \quad \bar{Q}_N = \begin{bmatrix} 0 \\ I_{N-P} \end{bmatrix}.$$

Let Π_1 and Π_2 represent row and column permutation matrices, respectively, defined as follows. For column permutations, we put at the beginning the first P columns of each occurrence of A_C , obtaining

$$\mathcal{J}\Pi_2 = \begin{bmatrix} R_1 & 0 & R_3 \\ R_2 & \mathbf{B}_{k_0, k_1} & R_4 \end{bmatrix}$$

where $R_1 = I_{k_1-k_0+1} \otimes A_C Q_N$

$$R_2 = \begin{bmatrix} W_{k_0} Q_N & -Q_N & 0_{N \times P} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0_{N \times P} & \dots & \dots & W_{k_1-1} Q_N & -Q_N \end{bmatrix}$$

$R_3 = I_{k_1-k_0+1} \otimes A_C \bar{Q}_N$, and

$$R_4 = \begin{bmatrix} W_{k_0} \bar{Q}_N & -\bar{Q}_N & 0 & \dots \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ 0 & \dots & W_{k_1-1} \bar{Q}_N & -\bar{Q}_N \end{bmatrix}.$$

For row permutations, consider the following steps: we first arrange the $(k_1 - k_0 + 1)$ row blocks corresponding to the first P rows of each occurrence of A_C , then the $(k_1 - k_0)$ row blocks corresponding to the first P rows of each occurrence of A_B , and, finally, the remaining rows, so as to obtain

$$\Pi_1 \mathcal{J} \Pi_2 = \begin{bmatrix} \mathbf{I}_{(k_1-k_0+1)P} & 0 & 0 \\ P_1 & \mathbf{I}_{(k_1-k_0)P} & P_2 \\ 0 & 0 & \tilde{\mathbf{C}} \\ P_3 & 0 & \tilde{\mathbf{W}} \end{bmatrix}$$

where

$$P_1 = \begin{bmatrix} \hat{W}_{k_0} & -I_P & 0 & \dots \\ \dots & \ddots & \ddots & \dots \\ 0 & \dots & \hat{W}_{k_1-1} & -I_P \end{bmatrix},$$

$$P_2 = \begin{bmatrix} \Lambda_{k_0} & 0 & \dots & \dots \\ \dots & \ddots & \dots & \dots \\ 0 & \dots & \Lambda_{k_1-1} & 0 \end{bmatrix},$$

$$P_3 = \begin{bmatrix} \Omega_{k_0} & 0 & \dots & \dots \\ \dots & \ddots & \dots & \dots \\ 0 & \dots & \Omega_{k_1-1} & 0 \end{bmatrix}$$

$\tilde{\mathbf{C}} = I_{k_1-k_0+1} \otimes \tilde{A}_C$, and

$$\tilde{\mathbf{W}} = \begin{bmatrix} \tilde{W}_{k_0} & -I_{N-P} & \dots & \dots & 0 \\ 0 & \tilde{W}_{k_0+1} & -I_{N-P} & \dots & 0 \\ \vdots & \dots & \dots & \dots & \vdots \\ 0 & \dots & \dots & \tilde{W}_{k_1-1} & -I_{N-P} \end{bmatrix}.$$

Let $\tilde{\mathcal{J}} = \Pi_1 \mathcal{J} \Pi_2$

$$\tilde{\mathcal{J}} = \begin{bmatrix} \mathbf{I}_{(k_1-k_0)P} & P_2 \\ 0 & \tilde{\mathbf{C}} \\ 0 & \tilde{\mathbf{W}} \end{bmatrix}$$

and

$$\tilde{\mathcal{J}} = \begin{bmatrix} \tilde{\mathbf{C}} \\ \tilde{\mathbf{W}} \end{bmatrix}.$$

Notice that $\tilde{\mathcal{J}}$ is block lower triangular with the blocks over the diagonal $\mathbf{I}_{(k_1-k_0+1)P}$ and $\tilde{\mathcal{J}}$. This implies $\text{rank}(\tilde{\mathcal{J}}) = (k_1 - k_0 + 1)P + \text{rank}(\tilde{\mathcal{J}})$. $\tilde{\mathcal{J}}$ is block upper triangular with blocks over the diagonal $\mathbf{I}_{(k_1-k_0)P}$ and $\tilde{\mathcal{J}}$. Therefore, the following holds:

$$\text{rank}(\tilde{\mathcal{J}}) = (k_1 - k_0 + 1)P + (k_1 - k_0)P + \text{rank}(\tilde{\mathcal{J}}).$$

From Proposition 4, we know that $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO over $[k_0, k_1]$ if and only if $\text{rank}(\tilde{\mathcal{J}}) = (k_1 - k_0)P + (k_1 - k_0 + 1)N$, which, in turn, is equivalent to $\text{rank}(\tilde{\mathcal{J}}) = (k_1 - k_0 + 1)(N - P)$. From Proposition 3, the latter corresponds to observability of $\{\tilde{W}_k, \tilde{A}_C\}_{k_0}^{k_1}$ over $[k_0, k_1]$. ■

The result in Proposition 5 can be interpreted as follows: when the subsystem $\{\tilde{W}_k, \tilde{A}_C\}_{k_0}^{k_1}$ is observable over $[k_0, k_1]$, one of the two unknown inputs in (3), namely, $\hat{x}_k$, is known and, hence, it is possible to compute u_k , since $\hat{x}_k$ is directly measured.

For LTI systems, alternatively, the PBH rank test may also be used to prove Proposition 5. The interested reader may be inspired by [20, Prop. 4] and the proof therein.

As mentioned previously, Propositions 1 and 4 characterize ISO in terms of rank conditions of matrices, namely, Ψ_{k_0, k_1} and $\mathcal{J}_{k_0, k_1}$, respectively. These algebraic techniques work well provided we have access to the exact values of all the coefficients of the aforementioned matrices. Moreover, from a computational standpoint, this technique is rather limited since the computational complexity increases as the size of the network grows. Therefore, in the sequel, we turn our attention to structural (respectively, s-structural) results, that is, the focus is on finding conditions such that the system is ISO for *almost all* choices of free parameters (respectively, *every* choice of free parameters) of the system matrices.

V. STRUCTURAL ISO

The main objective of this section is to characterize ISO for *almost all* choices of free parameters.

A. Definition and Implications

We denote by $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ the family of *all* LTV systems as given in (1) and having the same zero/nonzero pattern as given by $\mathcal{W}, A_B$, and A_C , while $\{\mathcal{W}, A_B, A_C\}_{\text{LTI}}$ represents the corresponding family of *all* LTI systems.

As mentioned previously, structured systems have fixed zero positions and free parameters. Let $|\mathcal{E}|$ denote the number of ones in A_G . Under such a setup, the free parameters can take values in $\mathbb{R}^{(|\mathcal{E}|)(k_1-k_0)}$, where $[k_0, k_1]$ represents the time window over which the system $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is being observed. Notice that each element in $\mathbb{R}^{(|\mathcal{E}|)(k_1-k_0)}$ yields a choice of free parameters. Structural ISO then asks that there be at least one member in $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ that is observable. This leads us to the following definition.

Definition 4: $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is *structurally ISO* on $[k_0, k_1]$, $k_1 > k_0$ if there exists *at least one* system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ with $W_k \in \mathcal{W}$, such that $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO. ■

Analogously, one can define structural observability and uniform N -step structural observability for LTV systems. In particular, definitions for structural ISO, structural observability, and uniform N -step structural observability can also be obtained for LTI systems where the space of free parameters is $\mathbb{R}^{|\mathcal{E}|}$ and the same free parameters are repeated at each time instant.

It is well known that observability (and controllability) is a property (see [29] and [12]) such that either there is no choice of parameters that makes it true, or it is true for almost all choices of parameters. Almost all choices of parameters mean all choices of the parameter except those lying in some proper algebraic variety of the space of free parameters. This means there are some nontrivial polynomials (one or more, but finitely many) such that the property is true for all parameters except those which are zeros of this system of polynomials. The polynomials being nontrivial (i.e., not identically zero) ensure that the variety is proper (i.e., not the whole space of free parameters) and, therefore, has Lebesgue measure zero. This can be interpreted as the property being true with probability one, if the parameters are thrown at random, according to any continuous probability distribution. Furthermore, small variations in the parameter values would not lead to loss of property. It turns out that the aforementioned discussion also holds for ISO, as shown in the following text:

Proposition 6: The set of parameters for which $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is not ISO is either the whole parameter space $\mathbb{R}^{(|\mathcal{E}|)(k_1-k_0)}$ or a proper variety of $\mathbb{R}^{(|\mathcal{E}|)(k_1-k_0)}$. ■

Proof: The proof is based on standard tools and is reported in the Appendix. ■

This means that over a given interval $[k_0, k_1]$, if one member of the family of systems $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is ISO, then *almost all* members of $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ are ISO. On the other hand, if $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is *not structurally* ISO, then *none* of the members of $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ are ISO. An analogous result to Proposition 6 also holds for LTI systems wherein the space of free parameters $\mathbb{R}^{|\mathcal{E}|}$. Definition 4 needs to be seen against this backdrop. It turns out that structural ISO for a family of LTI

systems implies structural ISO for the corresponding family of LTV systems and is given by the following remark.

Remark 3: If the LTI system $\{\mathcal{W}, A_B, A_C\}_{\text{LTI}}$ is structurally ISO, then the corresponding LTV system $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is structurally ISO over all sufficiently long intervals. Indeed, if the system $\{\mathcal{W}, A_B, A_C\}_{\text{LTI}}$ is structurally ISO, then there exists $W \in \mathcal{W}$, such that the triplet (W, A_B, A_C) is ISO. Therefore, over an interval $[k_0, k_1]$ of length at least N , one can set $W_k = W \quad \forall k \in [k_0, k_1]$, obtaining a system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ that is ISO over $[k_0, k_1]$, thereby exhibiting a choice of entries for which $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is ISO. Consequently, from Definition 4, the system $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is structurally ISO over $[k_0, k_1]$. ■

However, the converse of Remark 3 is open. In the remainder of this section, we show that, *under assumption A2*, the conditions given in Remark 3 are equivalent.

B. Uniform N -Step Structural ISO for LTV Systems

From Proposition 5, we can study ISO by studying the observability of a suitable subsystem. Here, we apply this technique to the family of systems $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$, defining a suitable family of subsystems. We define the set of matrices $\tilde{\mathcal{W}}$ as $\tilde{\mathcal{W}} = \{\tilde{Q}_N^T W \tilde{Q}_N \mid W \in \mathcal{W}\}$. Let $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTV}}$ represent the family of *all* LTV systems as given in (4) *but without the known input* $\hat{x}_k$. We denote by $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ the counterpart LTI subsystem (i.e., whose matrices have the same zero/nonzero pattern as given by $\tilde{\mathcal{W}}$). As a consequence of Proposition 5, for LTI systems, we have the following remark.

Remark 4: Under A2, $\{\mathcal{W}, A_B, A_C\}_{\text{LTI}}$ is uniform N -step structural ISO if and only if $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is structurally observable. ■

It turns out that corresponding to Remark 4, the conditions for *structural* results can also be obtained for LTV systems, as shall be evidenced in the remainder of this section.

An immediate corollary of Proposition 5 is the following.

Proposition 7: Under A2, $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is structurally ISO over $[k_0, k_1]$ if and only if $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTV}}$ is structurally observable over $[k_0, k_1]$. ■

The advantage of Proposition 7 is that it breaks down the problem of structural ISO into an equivalent problem in structural observability. With this in hand and rewriting [30, Th. 3] (also see [16]) for observability, we obtain equivalence between structural observability for LTV and LTI systems, and is given by the following proposition.

Proposition 8 (Theorem 3 in [30]): Under A2, over any interval $[k_0, k_1]$ of length at least N , $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTV}}$ is structurally observable if and only if $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is structurally observable. ■

Propositions 7 and 8 together break down the structural ISO problem of LTV systems into a structural observability problem of a corresponding suitably defined LTI subsystem. Thanks to [21], it turns out that the structural observability of an LTI subsystem can be determined by checking certain graph-theoretical conditions. Before proceeding, we need a few constructs on $\tilde{\mathcal{G}}$. Let $\tilde{\mathcal{G}}$ be the graph corresponding to $\tilde{\mathcal{W}}$. Let $\mathcal{S} = \{\mathcal{L}_1, \mathcal{L}_2, \mathcal{E}_S\}$ be a bipartite graph associated with $\tilde{\mathcal{G}}$,

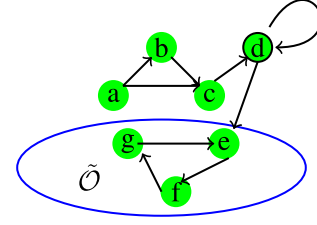


Fig. 2. Subsystem $\tilde{\mathcal{G}}$ for the system shown in Fig. 1.

with $\mathcal{L}_1 = \tilde{\mathcal{V}} \setminus \tilde{\mathcal{O}}$, $\mathcal{L}_2 = \tilde{\mathcal{V}}$ constructed in the following manner: two vertices in $\mathcal{L}_1$ and $\mathcal{L}_2$ that correspond to the same element $v \in \tilde{\mathcal{V}}$ are denoted as u_v and w_v , respectively, and $\mathcal{E}_S = \{(u_i, w_j) \in \mathcal{L}_1 \times \mathcal{L}_2 \mid (i, j) \in \tilde{\mathcal{E}}\}$. Similar to [31] but without the introduction of additional *output nodes* in $\tilde{\mathcal{G}}$, we state the following definitions.

Definition 5: The graph $\tilde{\mathcal{G}}$ with observation set $\tilde{\mathcal{O}}$ is said to be output-connected if for all $v \in \tilde{\mathcal{V}}$, there exists a path from v to w for some $w \in \tilde{\mathcal{O}}$. ■

Definition 6: A matching is a set of edges that do not share any common vertices. ■

With Definitions 5 and 6 in hand, we state the following result, rephrased for observability:

Lemma 1 (Theorem 1 [14]): The system $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is structurally observable if and only if

- 1) $\tilde{\mathcal{G}}$ is output-connected;
- 2) there exists a matching in $\mathcal{S}$ of size $N - |\tilde{\mathcal{O}}|$. ■

In addition, the aforementioned result previously appeared in [21] and [31]. With Lemma 1 in place, we state our first main result.

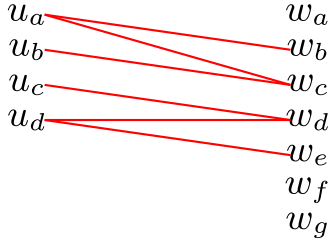
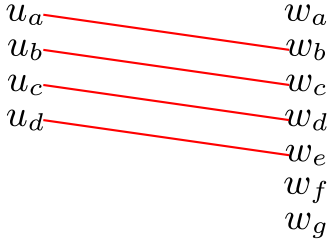
Theorem 1: Under A2, $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is uniformly N -step structurally ISO if and only if the following conditions are satisfied:

- 1) $\tilde{\mathcal{G}}$ is output-connected;
- 2) there exists a matching in $\mathcal{S}$ of size $N - |\tilde{\mathcal{O}}|$. ■

Proof: From Proposition 7, it can be seen that under A2, the system $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is structurally ISO over $[k_0, k_1]$ if and only if the subsystem $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTV}}$ is structurally observable over $[k_0, k_1]$, while from Proposition 8, it can be seen that the subsystem $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTV}}$ is structurally observable over $[k_0, k_1]$ if and only if the corresponding LTI subsystem $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is structurally observable. It is well known that LTI systems are either observable over *every* sufficiently long interval or *not* observable at all. Thus, setting $\delta = N$ in Remark 1, and from Propositions 7 and 8, it follows that under A2, the system $\{\mathcal{W}, A_B, A_C\}_{\text{LTV}}$ is structurally ISO over $[k_0, k_1]$ if and only if the subsystem $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is uniform N -step structural ISO. Thereafter, from Lemma 1, the proof is complete. ■

Example 1: With reference to the system given in Fig. 1, it can be seen from Figs. 2 and 4 that the subsystem is output-connected and its bipartite graph $\mathcal{S}$ contains a matching of size $N - |\tilde{\mathcal{O}}|$, and, hence, the subsystem is structurally observable [14]. Therefore, from Theorem 1, the system shown in Fig. 1 is uniformly N -step structurally ISO. ■

Item i) or output-connectedness of $\tilde{\mathcal{G}}$ can be checked by using a variant of Tarjan's algorithm and has complexity that is linear

Fig. 3. Bipartite graph $\mathcal{S}$ associated with $\tilde{\mathcal{G}}$.Fig. 4. Matching $\mathcal{M}$ in $\mathcal{S}$.

in the number of edges and vertices of $\tilde{\mathcal{G}}$ (i.e., $O(|\tilde{\mathcal{E}}| + |\tilde{\mathcal{V}}|)$) [32]. On the other hand, a Hopcroft–Karp maximum matching algorithm can be used for checking item ii) and its complexity is $O((|\tilde{\mathcal{E}}| + |\tilde{\mathcal{V}}|)\sqrt{|\tilde{\mathcal{V}}|})$ [33].

VI. S-STRUCTURAL ISO

The main objective of this section is to characterize ISO for every choice of entry in $\mathcal{W}^*$.

A. Definition

S-structural properties are those that hold for *every nonzero choice* of free parameters of the system matrices. That is, s-structural ISO (respectively, observability) requires that *every member* of the family of LTV systems given by $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ be ISO (respectively, observable). This leads us to the following definition.

Definition 7: Let $k_1, k_0 \in \mathbb{Z}$, and $k_1 > k_0$, $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ be *s-structurally ISO* on $[k_0, k_1]$ if for *every system* $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ with $W_k \in \mathcal{W}^*$, $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is ISO. ■

Analogous to Definition 7, one can also define *s-structural observability* and *uniform N-step s-structural observability* for LTV systems. In particular, definitions for *s-structural ISO*, *s-structural observability*, and *uniform N-step s-structural observability* can also be obtained for LTI systems.

It turns out that s-structural ISO for a family of LTV systems implies s-structural ISO for the corresponding family of LTI systems, and is given by the following remark:

Remark 5: If the system $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is s-structurally ISO over an interval $[k_0, k_1]$, then the corresponding LTI system $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTI}}$ is s-structurally ISO. Indeed, if the system $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTI}}$ is not s-structurally ISO, then from Definition 7, there exists a system $\{W_1, A_B, A_C\}$ with $W_1 \in \mathcal{W}^*$ such that $\{W_1, A_B, A_C\}$ is not ISO. Over any interval $[k_0, k_1]$, we can set $W_k = W_1 \quad \forall k \in [k_0, k_1]$ such that the LTV

system $\{W_k, A_B, A_C\}_{k_0}^{k_1}$ is not ISO. Consequently, from Definition 7, the system $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is not s-structurally ISO over any interval. ■

Notice that for structural ISO, the implication is in the other direction (see Remark 3).

The converse of Remark 5 remains open. In the remainder of this section, we show that, *under assumption A2*, over sufficiently long intervals, the conditions given in Remark 5 are equivalent.

B. Uniform N-Step S-Structural ISO for LTV Systems

The set of matrices $\tilde{\mathcal{W}}^*$ is defined analogous to $\tilde{\mathcal{W}}$. One can use Proposition 5 in order to obtain s-structural ISO results. We first focus on LTI systems. As another consequence of Proposition 5, we have the following.

Remark 6: Under A2, $\{\mathcal{W}, A_B, A_C\}_{\text{LTI}}$ is uniform N -step s-structural ISO if and only if $\{\tilde{\mathcal{W}}, \tilde{A}_C\}_{\text{LTI}}$ is s-structurally observable. ■

It turns out that one can obtain similar conditions for LTV systems as well, as shall be seen in the rest of this section.

First notice that another immediate corollary of Proposition 5 can be stated as follows.

Proposition 9: Under A2, $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is s-structurally ISO over $[k_0, k_1]$ if and only if $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTV}}$ is s-structurally observable over $[k_0, k_1]$. ■

Thanks to Proposition 9, we can now rephrase s-structural ISO of a family of LTV systems as an equivalent problem in s-structural observability of a suitable family of LTV systems. Against this backdrop, it is indeed relevant to see if the s-structural ISO of LTV systems is equivalent to s-structural observability of a suitable family of LTI systems. The following proposition is immediate from [17, Corollary IV.2].

Proposition 10: Under A2, over any interval $[k_0, k_1]$ of length at least N , $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTV}}$ is s-structurally observable if and only if $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTI}}$ is s-structurally observable. ■

Thus, from Propositions 9 and 10, it can be seen that under assumption A2, the s-structural ISO problem for LTV systems breaks down into an equivalent problem in s-structural observability for a suitably defined LTI subsystem. This equivalence allows us to exploit the literature on s-structural observability, as we see in the following text.

Thanks to [14] (also see [15]), it turns out that s-structural observability of an LTI system can be assessed by checking some graph-theoretical conditions. Here, we would be focusing on the notion of *uniquely restricted matching* (also known as constrained matching) as in [14]. In order to proceed, a few constructs on the graph $\tilde{\mathcal{G}}$ are due. $\mathcal{E}_{\text{loop}} \subset \mathcal{E}_{\mathcal{S}}$ denote the edges of the form $\{u_i, w_i\}$ if there exists any. Notice that $\mathcal{E}_{\text{loop}}$ corresponds to self-loops in $\tilde{\mathcal{G}}$. Let $\mathcal{E}_{\text{new}}$ denote the set of newly added self-loops in $\tilde{\mathcal{G}}$, that is, adding self-loops for those vertices $i \in \tilde{\mathcal{V}}$ that previously did not have one in $\tilde{\mathcal{G}}$. Let $\mathcal{S}_{\times} = \{\mathcal{L}_1, \mathcal{L}_2, \mathcal{E}_{\mathcal{S}_{\times}}\}$, where $\mathcal{E}_{\mathcal{S}_{\times}} = \{\mathcal{E}_{\mathcal{S}} \cup \mathcal{E}_{\text{new}}\}$, denote another bipartite graph on $\tilde{\mathcal{G}}$. We recall that a matching is said to be *uniquely restricted* if there is no other matching involving the same vertex set. Equivalent characterizations of uniquely restricted matchings are discussed

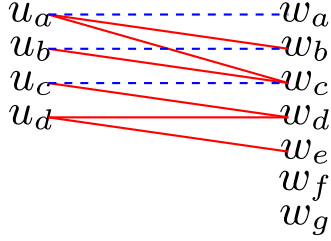


Fig. 5. Bipartite graph $\mathcal{S}_x$, in solid red: $\mathcal{S}_x \cap \mathcal{S}$, in dashed blue: $\mathcal{E}_{\text{new}} = \{(u_a, w_a), (u_b, w_b), (u_c, w_c)\}$, while $\mathcal{E}_{\text{loop}} = \{(u_d, w_d)\}$.

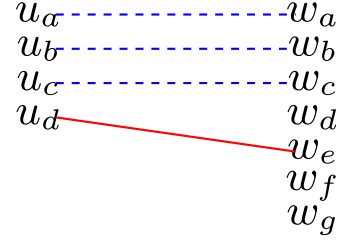


Fig. 6. Matching $\mathcal{M}_x$ in $\mathcal{S}_x$.

in [34]. The following result is the same as [14, Th. 5] but rewritten for s-structural observability.

Lemma 2 (Theorem 5 [14]): The system $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTI}}$ is s-structurally observable if and only if

- 1) there exists a uniquely restricted matching $\mathcal{M} \subseteq \mathcal{E}_S$ of size $N - |\tilde{\mathcal{O}}|$;
- 2) there exists a uniquely restricted matching $\mathcal{M}_x \subseteq \mathcal{E}_{S_x}$ of size $N - |\tilde{\mathcal{O}}|$ such that $\mathcal{M}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$. ■

With Lemma 2 in place, we present our second main result in the following theorem.

Theorem 2: Under A2, $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is uniformly N -step s-structurally ISO if and only if the following two conditions are satisfied:

- 1) there exists a uniquely restricted matching $\mathcal{M} \subseteq \mathcal{E}_S$ of size $N - |\tilde{\mathcal{O}}|$;
- 2) there exists a uniquely restricted matching $\mathcal{M}_x \subseteq \mathcal{E}_{S_x}$ of size $N - |\tilde{\mathcal{O}}|$ such that $\mathcal{M}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$. ■

Proof: From Propositions 9 and 10, it can be seen that under A2, $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is s-structurally ISO over any interval of length at least N if and only if $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTI}}$ is s-structurally observable, while from Proposition 10, it can also be seen that $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTI}}$ is s-structurally observable if and only if $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTV}}$ is s-structurally observable over *every* interval of length at least N . Thus, Propositions 9 and 10, together with setting $\delta = N$ in Remark 1, result in the following: under A2, $\{\mathcal{W}^*, A_B, A_C\}_{\text{LTV}}$ is s-structurally ISO if and only if the LTI subsystem $\{\tilde{\mathcal{W}}^*, \tilde{A}_C\}_{\text{LTI}}$ is uniform N -step s-structural ISO. Thereafter, from Lemma 2, the proof is complete. ■

The conditions in Theorem 2 can be checked using the algorithm given in [14], with complexity $O(|\tilde{\mathcal{V}}|^2)$, or with the algorithm introduced in [35], which achieves a linear complexity $O(|\tilde{\mathcal{V}}| + |\tilde{\mathcal{E}}|)$ by combining sophisticated data structures and sparse matrix techniques.

Example 1 (continued): With respect to the system shown in Fig. 1, first recall that $\mathcal{M} = \{(u_a, w_b), (u_b, w_c), (u_c, w_d), (u_d, w_e)\}$ (see Fig. 4) is a matching in $\mathcal{S}$. Furthermore, there exists no other matching $\tilde{\mathcal{M}} \subset \mathcal{E}_S$ saturating the same vertices as $\mathcal{M}$ and, hence, by definition, $\mathcal{M} \subset \mathcal{E}_S$ is a uniquely restricted matching. The second condition is checked with respect to the bipartite graph $\mathcal{S}_x$ shown in Fig. 5.

It can be seen that $\mathcal{M}_x = \{(u_a, w_a), (u_b, w_b), (u_c, w_c), (u_d, w_e)\}$ (see Fig. 6) is a matching in $\mathcal{S}_x$ that satisfies $\mathcal{M}_x \cap \mathcal{E}_{\text{loop}} = \emptyset$. Notice that there exists no other matching $\hat{\mathcal{M}}_x \subset \mathcal{E}_{S_x}$ saturating the same vertices as $\mathcal{M}_x$. Therefore, from

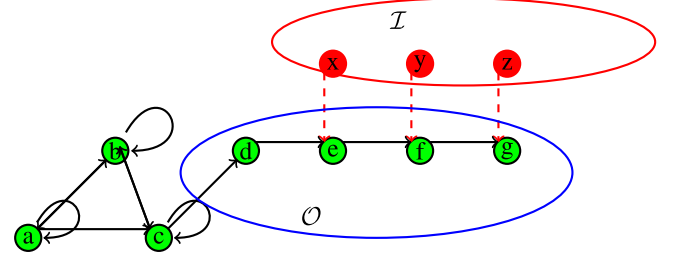


Fig. 7. System $\mathcal{G}_1$.

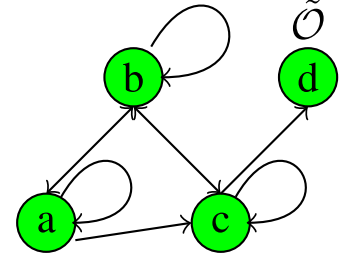


Fig. 8. Subsystem $\tilde{\mathcal{G}}_1$.

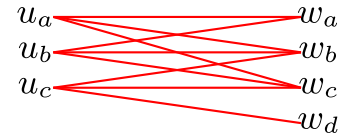


Fig. 9. Bipartite graph $\mathcal{S}_1$ associated with $\tilde{\mathcal{G}}_1$.

Theorem 2, the system shown in Fig. 1 is uniformly N -step s-structurally ISO. ■

Now consider another example.

Example 2: Consider the system shown in Fig. 7, whose corresponding subsystem is shown in Fig. 8, while a bipartite graph associated with the subsystem $\tilde{\mathcal{G}}_1$ is shown in Fig. 9. It is immediate that the subsystem $\tilde{\mathcal{G}}_1$ is output-connected. Furthermore, there also exists a matching of size $N - |\tilde{\mathcal{O}}|$ on the bipartite graph $\mathcal{S}_1$. Therefore, from Theorem 1, the system shown in Fig. 7 is uniformly N -step structurally ISO. On the other hand, from Fig. 10(a) and (b), it can be seen that there does not exist a uniquely restricted matching over the choice of vertex sets $\{u_a, u_b, u_c\}$ and $\{w_a, w_b, w_c\}$. The same can be said with respect to the vertex sets $\{u_a, u_b, u_c\}$ and $\{w_a, w_b, w_d\}$ [see Fig. 10(c) and (d)], $\{u_a, u_b, u_c\}$ and $\{w_b, w_c, w_d\}$ [see Fig. 10(e) and (f)], and $\{u_a, u_b, u_c\}$ and $\{w_a, w_c, w_d\}$ [see Fig. 10(g) and

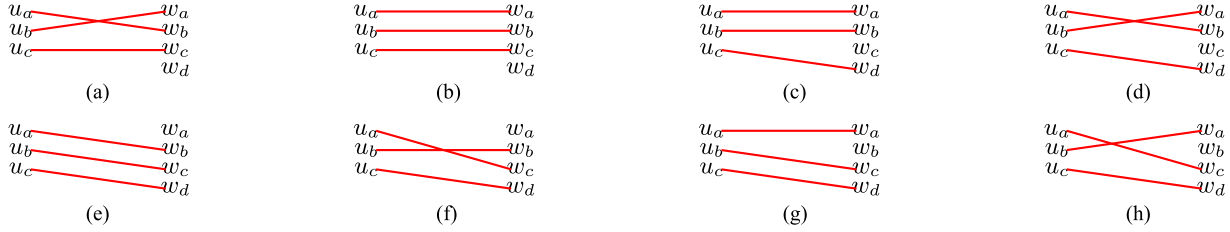


Fig. 10. All maximum matchings on S_1 . (a) Matching $\mathcal{M}_1$ on S_1 . (b) Matching $\mathcal{M}_2$ on S_1 . (c) Matching $\mathcal{M}_3$ on S_1 . (d) Matching $\mathcal{M}_4$ on S_1 . (e) Matching $\mathcal{M}_5$ on S_1 . (f) Matching $\mathcal{M}_6$ on S_1 . (g) Matching $\mathcal{M}_7$ on S_1 . (h) Matching $\mathcal{M}_8$ on S_1 .

(h)]. Thus, there does not exist a uniquely restricted matching of size $N - |\tilde{\mathcal{O}}|$ on the bipartite graph S_1 and, hence, from Theorem 2, the system shown in Fig. 7 is *not* uniformly N -step s-structurally ISO. ■

VII. CONCLUSION

We have studied ISO with a one time-step delay for linear network systems with a fixed topology. Under the assumptions 1) each assailable node can be attacked by, at most, a single unknown input; 2) each unknown input affects exactly one node; and 3) direct measurements of certain states are available, we provide a characterization of ISO in terms of observability of a suitably defined subsystem. Moreover, we have studied uniform N -step ISO for *almost all* choices of free parameters in $\mathcal{W}$ as well as for *all nonzero* choices of free parameters in $\mathcal{W}$, where $\mathcal{W}$ stands for the family of systems sharing the same zero/nonzero pattern. A future line of investigation would be to characterize ISO for more general linear network systems by considering time-varying topology and less *a priori* knowledge on the assailable nodes. A weaker notion of ISO, wherein the focus is on the unique recovery of just a subset of the states, for instance, only the states that are *not* affected by an unknown input, could be another direction of future works.

APPENDIX

A. Proof of Proposition 2

Item i) requires that the first N columns of Ψ_{k_0, k_1} be linearly independent. Item ii) requires that the last P columns of Ψ_{k_0, k_1} be linearly independent, while items iii) and iv) are necessary conditions for item ii). To see the necessity of items v) and vi), notice that in order for Ψ_{k_0, k_1} to be full-column rank, it is necessary that Ψ_{k_0, k_1} has at least as many rows as columns, i.e.,

$$M(k_1 - k_0 + 1) \geq N + (k_1 - k_0)P.$$

From the above equation, since $(k_1 - k_0 + 1) > 0$, it follows that $M \geq P + \frac{N-P}{(k_1-k_0+1)}$. If $N > P$, this implies that $M > P$. Then, under $M > P$, item vi) immediately follows from the above equation.

For the particular case of $M = P = N$, notice that Ψ_{k_0, k_1} is a block lower triangular matrix with each of the blocks being square. Hence, a necessary and sufficient condition for full-column rank of Ψ_{k_0, k_1} is that each of the diagonal

blocks has full-column rank. This is equivalent to 1) $\text{rank}(A_C A_B) = N \quad \forall k \in [k_0 + 1, k_1]$ and 2) $\text{rank}(A_C) = N$. Notice that $\forall k \in [k_0 + 1, k_1]$, $\text{rank}(A_C A_B) = N$ if and only if 1) $\text{rank}(A_C) = N$ and 2) $\text{rank}(A_B) = N$. ■

B. Proof of Proposition 6

From Proposition 1, ISO is equivalent to Ψ_{k_0, k_1} having rank $N + (k_1 - k_0)P$. Hence, the system is not ISO if and only if all of the square submatrices of size $N + (k_1 - k_0)P$ have a zero determinant. Notice that the entries of Ψ_{k_0, k_1} are polynomials, whose variables are the free parameters of W_k , A_B , and A_C ; the fixed values (zeros) can be interpreted as polynomials of degree zero. For each submatrix of size $N + (k_1 - k_0)P$, the determinant is obtained by multiplications and summations of such polynomials and, hence, is itself a polynomial. We have found a finite set of polynomials, such that the system is not ISO if and only if the parameters belong to the zero set of all these polynomials. Either such polynomials are all trivial (i.e., constantly equal to zero) and, hence, all choices of parameters result in a system not ISO, or at least one of the polynomials is nontrivial and, hence, the set of parameters for which the system is not ISO is a proper variety of parameter space. ■

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